



Geography: Global

Astronomy - v3

Domain Portfolio: Forecast | Domain: Daily Forecasts | Usage Classification: Standard

Attribution Required: No

Attribution Requirements: N/A

Overview

The Astronomy API is a public API sourced from the The Weather Company Forecast system. The astronomy information is calculated based on the location's latitude and longitude. This API returns astronomy information pertaining to sun rise and set, moon rise and set and moon phases.

Atomic Endpoints	Aggregate Product Names
v3/wx/forecast/astronomy/daily/1day	v3-wx-forecast-astronomy-daily-1day
v3/wx/forecast/astronomy/daily/5day	v3-wx-forecast-astronomy-daily-5day
v3/wx/forecast/astronomy/daily/7day	v3-wx-forecast-astronomy-daily-7day
v3/wx/forecast/astronomy/daily/15day	v3-wx-forecast-astronomy-daily-15day
v3/wx/forecast/astronomy/daily/30day	v3-wx-forecast-astronomy-daily-30day

Translated Fields:

This TWC API handles the translation of phrases. However, when formatting a request URL a valid language must be passed along (see the language code table for the supported codes).

- lunarPhaseDescription

URL Construction

Request by Geocode: Required Parameters: geocode, units, language, format, day, month, year (for 1day), startDay, startMonth, startYear (for all other days), apiKey https://api.weather.com/v3/wx/forecast/astronomy/daily/1day?geocode=<geocode>&language=<language>&format=<format>&units=<units>&day=<day>&month=<month>&year=<year>&apiKey=yourApiKey https://api.weather.com/v3/wx/forecast/astronomy/daily/7day?geocode=<geocode>&language=<language>&format=<format>&units=<units>&startDay=<startDay>&startMonth=<startMonth>&startYear=<startYear>&apiKey=yourApiKey
https://api.weather.com/v3/wx/forecast/astronomy/daily/1day?geocode=34,-84&language=en-US&format=json&units=h&day=01&month=12&year=2016&apiKey=yourApiKey https://api.weather.com/v3/wx/forecast/astronomy/daily/7day?geocode=34,-84&language=en-US&format=json&units=h&startDay=01&startMonth=12&startYear=2016&apiKey=yourApiKey
Request by IATA Code: Required Parameters: iataCode, units, language, format, day, month, year (for 1day), startDay, startMonth, startYear (for all other days), apiKey https://api.weather.com/v3/wx/forecast/astronomy/daily/1day?iataCode=<iataCode>&language=<language>&format=<format>&units=<units>&day=<day>&month=<month>&year=<year>&apiKey=yourApiKey https://api.weather.com/v3/wx/forecast/astronomy/daily/7day?iataCode=<iataCode>&language=<language>&format=<format>&units=<units>&startDay=<startDay>&startMonth=<startMonth>&startYear=<startYear>&apiKey=yourApiKey
https://api.weather.com/v3/wx/forecast/astronomy/daily/1day?iataCode=DEN&language=en-US&format=json&units=h&day=01&month=12&year=2016&apiKey=yourApiKey https://api.weather.com/v3/wx/forecast/astronomy/daily/7day?iataCode=DEN&language=en-US&format=json&units=h&startDay=01&startMonth=12&startYear=2016&apiKey=yourApiKey
Request by ICAO Code: Required Parameters: icaoCode, units, language, format, day, month, year (for 1day), startDay, startMonth, startYear (for all other days), apiKey https://api.weather.com/v3/wx/forecast/astronomy/daily/1day?icaoCode=<icaoCode>&language=<language>&format=<format>&units=<units>&day=<day>&month=<month>&year=<year>&apiKey=yourApiKey

https://api.weather.com/v3/wx/forecast/astronomy/daily/7day?icaoCode=<icaoCode>&language=<language>&format=<format>&units=<units>&startDay=<startDay>&startMonth=<startMonth>&startYear=<startYear>&apiKey=yourApiKey
https://api.weather.com/v3/wx/forecast/astronomy/daily/1day?icaoCode=KDEN&language=en-US&format=json&units=h&day=01&month=12&year=2016&apiKey=yourApiKey https://api.weather.com/v3/wx/forecast/astronomy/daily/7day?icaoCode=KDEN&language=en-US&format=json&units=h&startDay=01&startMonth=12&startYear=2016&apiKey=yourApiKey
Request by Place ID: Required Parameters: placeid, units, language, format, day, month, year (for 1day), startDay, startMonth, startYear (for all other days), apiKey https://api.weather.com/v3/wx/forecast/astronomy/daily/1day?placeid=<placeid>&language=<language>&format=<format>&units=<units>&day=<day>&month=<month>&year=<year>&apiKey=yourApiKey https://api.weather.com/v3/wx/forecast/astronomy/daily/7day?placeid=<placeid>&language=<language>&format=<format>&units=<units>&startDay=<startDay>&startMonth=<startMonth>&startYear=<startYear>&apiKey=yourApiKey
https://api.weather.com/v3/wx/forecast/astronomy/daily/1day?placeid=ee0214ae7fb1d265f3e4d2509e77557119b2cac9d0ac1013b593c74a1567f2b7&language=en-US&format=json&units=h&day=01&month=12&year=2016&apiKey=yourApiKey https://api.weather.com/v3/wx/forecast/astronomy/daily/7day?placeid=ee0214ae7fb1d265f3e4d2509e77557119b2cac9d0ac1013b593c74a1567f2b7&language=en-US&format=json&units=h&startDay=01&startMonth=12&startYear=2016&apiKey=yourApiKey
Request by Postal Key: Required Parameters: postalKey, units, language, format, day, month, year (for 1day), startDay, startMonth, startYear (for all other days), apiKey https://api.weather.com/v3/wx/forecast/astronomy/daily/1day?postalKey=<postalKey>&language=<language>&format=<format>&units=<units>&day=<day>&month=<month>&year=<year>&apiKey=yourApiKey https://api.weather.com/v3/wx/forecast/astronomy/daily/7day?postalKey=<postalKey>&language=<language>&format=<format>&units=<units>&startDay=<startDay>&startMonth=<startMonth>&startYear=<startYear>&apiKey=yourApiKey
https://api.weather.com/v3/wx/forecast/astronomy/daily/1day?postalKey=81657:US&language=en-US&format=json&units=h&day=01&month=12&year=2016&apiKey=yourApiKey https://api.weather.com/v3/wx/forecast/astronomy/daily/7day?postalKey=81657:US&language=en-US&format=json&units=h&startDay=01&startMonth=12&startYear=2016&apiKey=yourApiKey

Valid Parameter Definitions

Parameter Name	Valid Parameter Value	Description
day	ISO DD	ISO day specified for astronomy daily forecasts 1 Day data return.
startDay	ISO DD	ISO day specified for astronomy daily forecasts 7 Day data return.
month	ISO MM	ISO month specified for astronomy daily forecasts 1 Day data return.
startMonth	ISO MM	ISO month specified for astronomy daily forecasts 7 Day data return.
year	ISO YYYY	ISO year specified for astronomy daily forecasts 1 Day data return.
startYear	ISO YYYY	ISO year specified for astronomy daily forecasts 7 Day data return.

Data Elements & Definitions

Note: Field names are sorted alphabetically in the table below for presentation purposes. The table below does not represent the sort order of the API response.

Field Name	Description	Type	Range	Sample	Nulls
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					Allowed
astronomicalDawnTimeLocal	Astronomical Dawn in ISO local time	[ISO]	-	"2016-01-01T06:11:57.349-05:00"	Y
astronomicalDawnTimeUtc	Astronomical Dawn in epoch UNIX time	[epoch]	-	1451646717	Y
astronomicalDuskTimeLocal	Astronomical Dusk in ISO local time	[ISO]	-	"2016-01-01T19:09:05.820-05:00"	Y
astronomicalDuskTimeUtc	Astronomical Dusk in epoch UNIX time	[epoch]	-	1451693345	Y
civilDawnTimeLocal	Sunrise time in ISO local time	[ISO]	-	"2016-01-01T07:12:51.329-05:00"	Y
civilDawnTimeUtc	Sunrise time in epoch UNIX time	[epcoh]	-	1451650371	Y
civilDuskTimeLocal	Sunset time in ISO local time	[ISO]	-	"2016-01-01T18:08:11.840-05:00"	Y
civilDuskTimeUtc	Sunrise time in epoch UNIX time	[epcoh]	-	1451689691	Y
dayofWeek	Day of week	[string]	-	"FRIDAY"	N
lunarAge	Age of the moon in days.	[decimal]	0-29.5	21.08495266	Y
lunarPhaseCode	Numeric 1-8 mapping of the lunar phase. NewMoon = 1, WaxingCrescent = 2, FirstQuarter = 3, WaxingGibbous = 4, FullMoon = 5, WaningGibbous = 6, LastQuarter = 7, WaningCrescent = 8	[integer]	1-8	6	Y
lunarPhaseDescription	Moon phase description. Note that lunarAge to lunarPhaseDescription is not static.	[string]	New Moon, Waxing Crescent, First Quarter, Waxing Gibbous, Full Moon, Waning Gibbous, Last Quarter, Waning Crescent	"Waning Gibbous"	Y
moonIllumination	Percentage of the moon illuminated	[decimal]	0-100	4.05	Y
moonriseIconCode	Icon number corresponding to the moon phase icon image at moonset	[integer]	0-29	21	Y
moonsetIconCode	Icon number corresponding to the moon phase icon image at moonrise	[integer]	1-30	22	Y
moonriseTimeLocal	Moonrise time in local time. It reflects daylight savings time conventions.	[ISO]	-	"2016-01-01T00:51:00.000-05:00"	Y
moonsetTimeLocal	Moonset time in local time. It reflects daylight savings time conventions.	[ISO]	-	"2016-01-01T13:09:00.000-05:00"	Y
moonriseTimeUtc	Moonrise time in epoch UNIX time. It reflects daylight savings time conventions.	[epoch]	-	1451627460	Y
moonsetTimeUtc	Moonset time in epoch UNIX time. It reflects daylight savings time conventions.	[epoch]	-	1451671740	Y
nauticalDawnTimeLocal	Nautical dawn time in local time	[ISO]	-	"2016-01-01T06:42:03.053-05:00"	Y
nauticalDawnTimeUtc	Nautical dawn time in epoch UNIX time	[epoch]	-	1451648523	Y
nauticalDuskTimeLocal	Nautical dusk time in local time	[ISO]	-	"2016-01-01T18:39:00.116-05:00"	Y
nauticalDuskTimeUtc	Nautical dusk in epoch UNIX time	[epoch]	-	1451691540	Y
sunriseTimeLocal	Sunrise time in local time. It reflects daylight savings time conventions.	[ISO]	-	"2016-01-01T07:40:08.344-05:00"	Y
sunriseTimeUtc	Sunrise time in epoch UNIX time. It reflects daylight savings time conventions.	[epoch]	-	1451652008	Y
sunsetTimeLocal	Sunset time in local time. It reflects daylight savings time conventions.	[ISO]	-	"2016-01-01T17:40:54.825-05:00"	Y
sunsetTimeUtc	Sunset time in epoch UNIX time. It reflects daylight savings time conventions.	[epoch]	-	1451688054	Y
zenithTimeLocal	Zenith time in local time. It reflects daylight savings time conventions.	[ISO]	-	"2016-01-01T12:40:31.584-05:00"	Y
zenithTimeUtc	Zenith time in epoch UNIX time. It reflects daylight savings time conventions.	[epoch]	-	1451670031	Y

JSON Sample

```
{
  astronomicalDawnTimeLocal:
  [
    "2016-01-01T06:11:57.349-05:00"
  ],
  astronomicalDawnTimeUtc:
  [
    1451646717
  ],
  astronomicalDuskTimeLocal:
  [
    "2016-01-01T19:09:05.820-05:00"
  ],
  astronomicalDuskTimeUtc:
  [
    1451693345
  ],
  civilDawnTimeLocal:
  [
    "2016-01-01T07:12:51.329-05:00"
  ],
  civilDawnTimeUtc:
  [
    1451650371
  ],
  civilDuskTimeLocal:
  [
    "2016-01-01T18:08:11.840-05:00"
  ],
  civilDuskTimeUtc:
  [
    1451689691
  ],
  dayOfWeek:
  [
    "FRIDAY"
  ],
  lunarAge:
  [
    21.084952663558596
  ],
  lunarPhaseCode:
```

```
[
  6
],
lunarPhaseDescription:
[
  "Waning Gibbous"
],
moonIllumination:
[
  4.05
],
moonriseIconCode:
[
  21
],
moonsetIconCode:
[
  22
],
moonriseTimeLocal:
[
  "2016-01-01T00:51:00.000-05:00"
],
moonsetTimeLocal:
[
  "2016-01-01T13:09:00.000-05:00"
],
moonriseTimeUTC:
[
  1451627460
],
moonsetTimeUTC:
[
  1451671740
],
nauticalDawnTimeLocal:
[
  "2016-01-01T06:42:03.053-05:00"
],
nauticalDawnTimeUTC:
[
  1451648523
],
```

```
    nauticalDuskTimeLocal:
    [
        "2016-01-01T18:39:00.116-05:00"
    ],
    nauticalDuskTimeUTC:
    [
        1451691540
    ],
    sunriseTimeLocal:
    [
        "2016-01-01T07:40:08.344-05:00"
    ],
    sunriseTimeUTC:
    [
        1451652008
    ],
    sunsetTimeLocal:
    [
        "2016-01-01T17:40:54.825-05:00"
    ],
    sunsetTimeUTC:
    [
        1451688054
    ],
    zenithTimeLocal:
    [
        "2016-01-01T12:40:31.584-05:00"
    ],
    zenithTimeUTC:
    [
        1451670031
    ]
}
```